ME5103 Finite Element Methods
Lab 5 – 2D elements CST, QUAD

30/10/2026

Use an FEM software of your choice or ANSYS APDL.

- *TRI and QUAD (higher order if needed) elements can be used to mesh the domain if you prefer 2D analysis. E = 200 GPa and v = 0.3. Use different types of 2D meshes for mesh sensitivity study.*
- *Prepare a report showing the problem formulation in the Cartesian coordinate system.*
- *The results should contain the stress plots and a table comparing the exact solution and the FEM solution.*
- *Carry out a mesh sensitivity study (i.e., do the same analysis for different mesh sizes until you get a converged results)*
- *The conclusions and inferences section should present the key results, outline the challenges faced, and describe the measures taken to overcome them in achieving the final solution.*

1. A plate of length a = 300 mm and height b = 300 mm has a hole of radius 10 mm. The plate is subjected to pure shear traction 100 MPa at $x = \pm 150$ mm and $y = \pm 150$ mm.
    a. Solve for the in-plane stresses in the entire domain. Find out the ***maximum shear stress*** around the hole and find out the stress concentration factor.
    b. Extract the contour plots of the stresses and comment on the results.
    c. Consider a section A-A along y = 0 and plot the variation of the shear stress along A-A.
    d. Compare the results with the solution discussed in class using Airy's stress function. Comment on the results.

    [5 Marks]

2. The plate in the above problem is subjected to uniaxial tension T = 100 MPa, at $x = \pm 150$ mm, instead of shear traction S.
    a. Use only $1/4^{th}$ of the geometry to model the problem and apply symmetry boundary conditions at x=0 and y=0.
    b. Solve for the in-plane stresses in the entire domain. Find out the ***maximum normal stress*** around the hole and find out the stress concentration factor.
    c. Extract the contour plots of the stresses and comment on the results.
    d. Consider a section A-A along x =0, y = 0 and plot the variation of the shear stress along A-A.
    e. Compare the results with the solution with exact solutions from literature. Comment on the results.

    [5 Marks]